

> 7.2 Gametes and inheritance

In this topic you will:

- learn what a gamete is
- think about the similarities and differences between male and female gametes
- find out what happens at fertilisation
- explain what determines whether a baby is a boy or a girl.

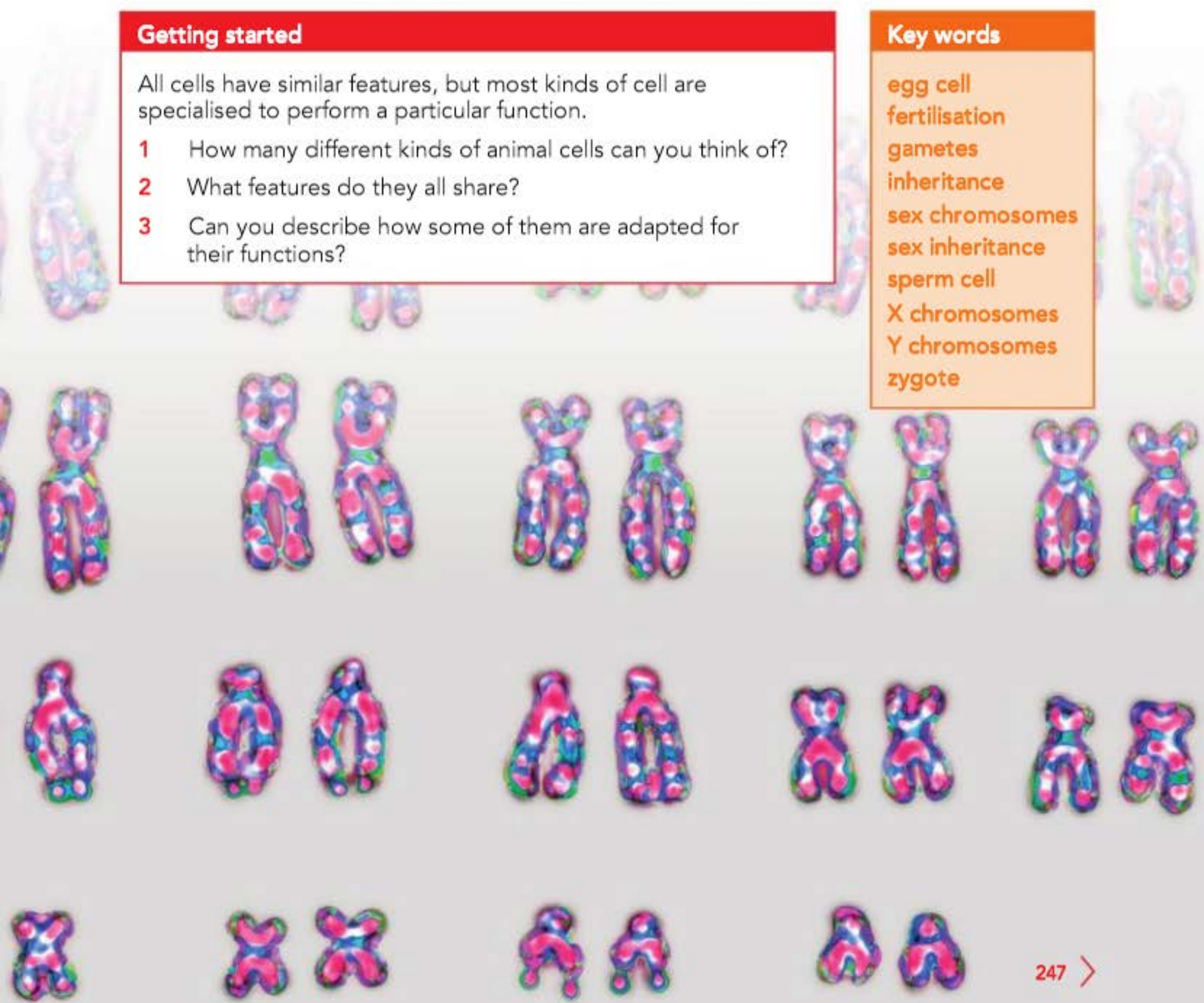
Getting started

All cells have similar features, but most kinds of cell are specialised to perform a particular function.

- 1 How many different kinds of animal cells can you think of?
- 2 What features do they all share?
- 3 Can you describe how some of them are adapted for their functions?

Key words

egg cell
fertilisation
gametes
inheritance
sex chromosomes
sex inheritance
sperm cell
X chromosomes
Y chromosomes
zygote

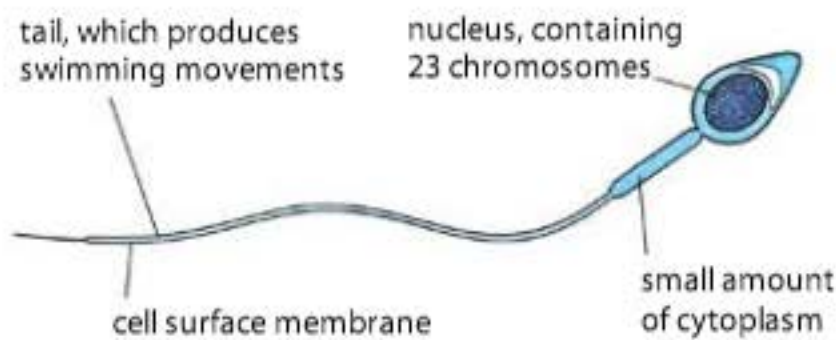


Gametes

Most cells in a human body contain 46 chromosomes. Where did these chromosomes come from? To answer that, try to think about how a new human life begins.

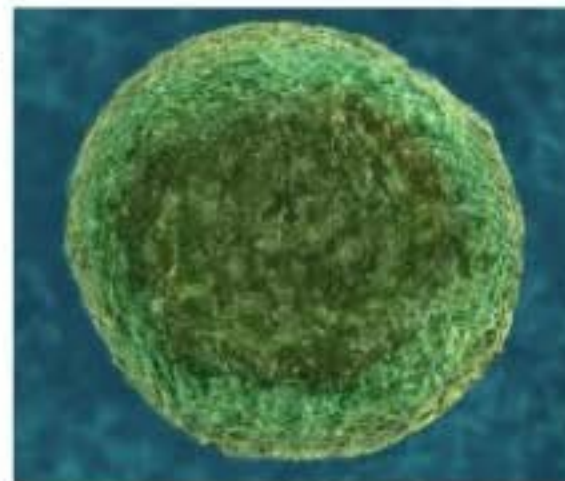
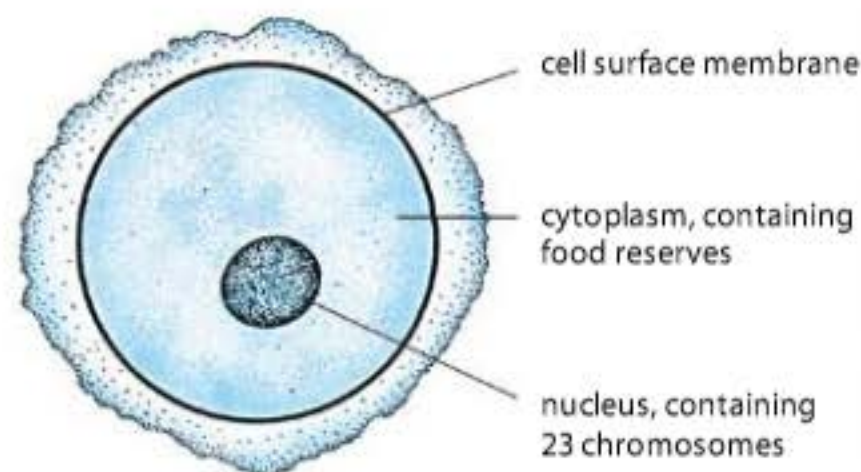
Every human being began life as a single cell. This cell was formed when a **sperm cell** joined with an **egg cell**. Sperm cells and egg cells are specialised cells known as **gametes**. A sperm cell is a male gamete, and an egg cell is a female gamete.

This diagram shows a human sperm cell. The photograph shows a group of sperm cells. It was taken with a special kind of powerful microscope called a scanning electron microscope.



Sperm cells are very small cells. They are very active, using their tails to swim vigorously.

This diagram shows a human egg cell. The photograph, like the photograph of the sperm cells, was taken with a scanning electron microscope.



Egg cells are much bigger than sperm cells, but they are still very small. A human egg cell is about the same size as a full stop. They need to be larger than sperm cells because they contain food reserves. Another difference from sperm cells is that egg cells cannot move by themselves.

Questions

- 1 What is a gamete?
- 2 Draw a table with two columns. Write the headings 'Sperm cell' and 'Egg cell' at the top of the columns. Then make a comparison of these two cells. Try to think of at least three differences between them.

Fertilisation

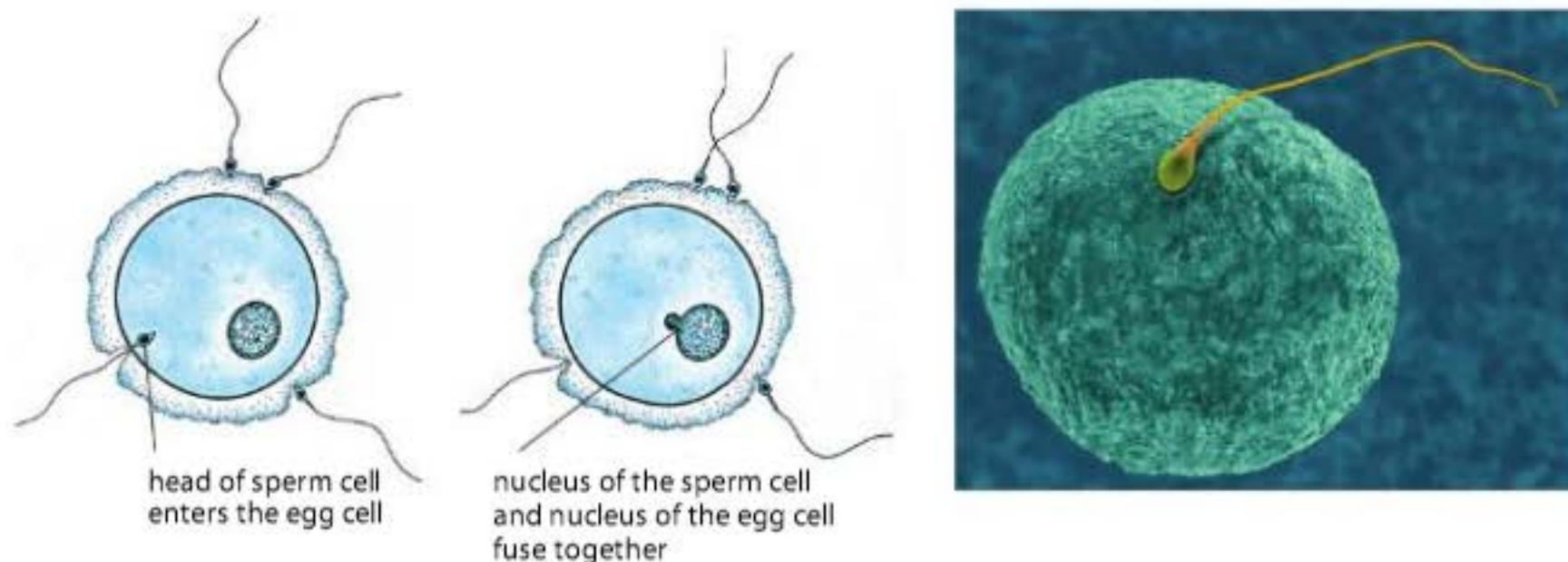
You may have noticed something odd in the labels on the diagrams of the sperm cell and the egg cell. They each have only 23 chromosomes.

In the previous topic, we saw that human cells each have 46 chromosomes in their nucleus. They have two sets, each set containing 23 chromosomes.

But gametes have only one set of chromosomes. This means that, when a sperm cell joins with an egg cell, the new cell that is produced has two sets. It will have 46 chromosomes.

The joining of a sperm cell with an egg cell is called **fertilisation**.

The diagram and photograph show how fertilisation takes place.



The new cell that is formed when the nucleus of the sperm cell and the egg cell join together is called a **zygote**.

All humans began their life as a single cell like this. Over the next days, weeks and years, this single cell divides over and over again, eventually producing all of the millions of cells in a human body.

Boy or girl?

All of the chromosomes in the cells in a human body came from the person's mother and father. There is one set from the mother, and one set from the father. We can use this information to explain how a baby's sex is determined.

Look at the black and white photograph in the previous topic, where the two sets of chromosomes in a cell from a woman's body have been arranged in order. The last pair, at the bottom right of the photograph, are labelled X.

These two chromosomes are **sex chromosomes**. They determine whether a person is male or female. In that photograph, there are two **X chromosomes**. A person with two X chromosomes, XX, is female.

There is another kind of sex chromosome, called a **Y chromosome**.

Y chromosomes are much smaller than X chromosomes. A person with one X chromosome and one Y chromosome, XY, is male.

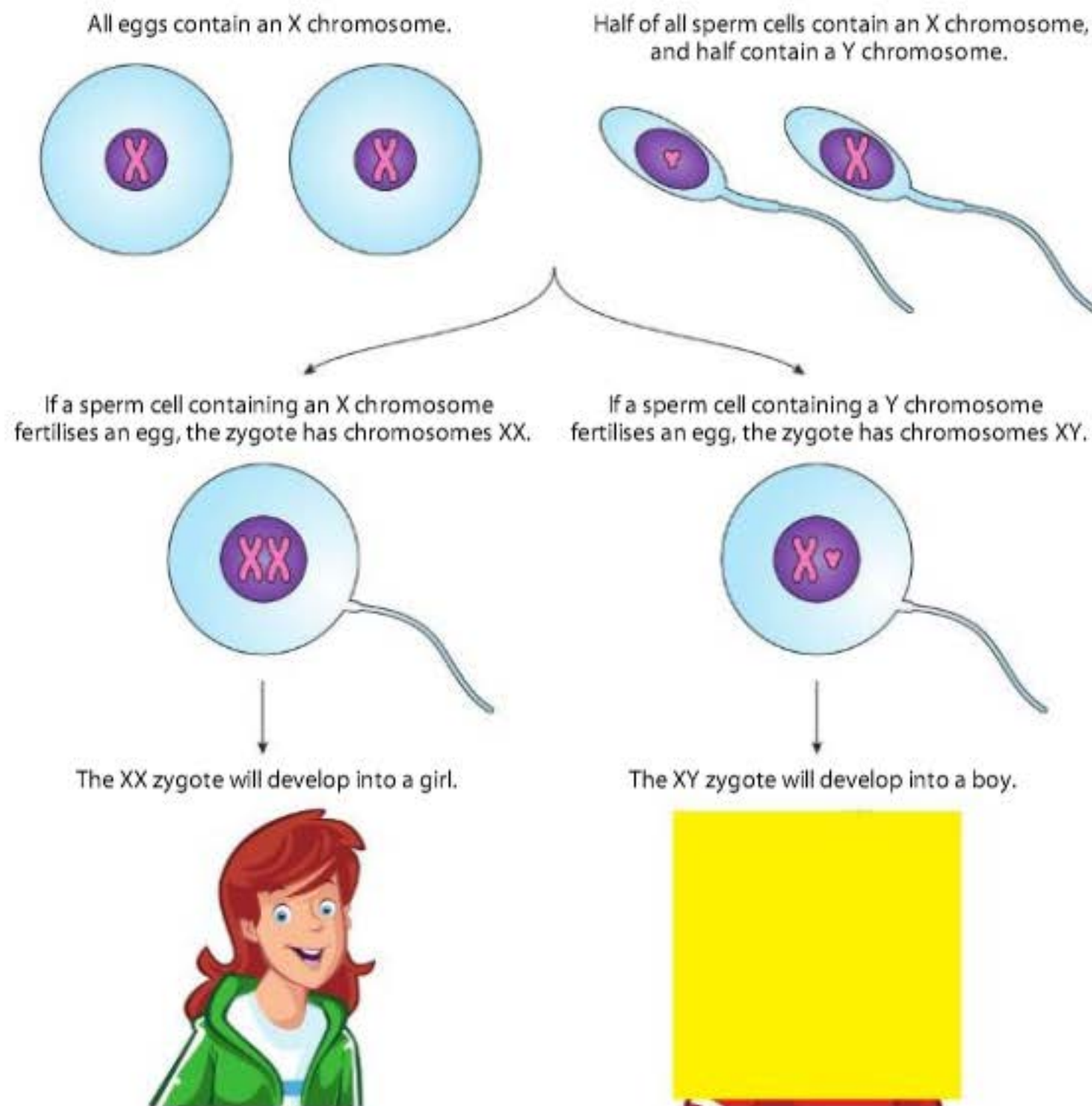
This photograph – again taken with a scanning electron microscope – shows a human X chromosome and Y chromosome.



All egg cells contain one X chromosome. Remember that gametes only have one set of chromosomes, so they only have one of each kind. However, sperm cells can have either one X chromosome or one Y chromosome.

So, each time fertilisation happens, either an X-containing sperm cell or a Y-containing sperm cell could join with an egg. If it is an X-containing sperm cell, then the zygote will have two X chromosomes, and will become a baby girl. If it is a Y-containing sperm cell, then the zygote will have one X chromosome and one Y chromosome, and will be a baby boy.

The chance of either of these events happening is about equal. This is why approximately equal numbers of boys and girls are born.



Inheritance

Inheritance means passing on DNA (as chromosomes containing genes) from parents to offspring.

A baby's sex is determined because a baby inherits X or Y chromosomes from its parents. We can use the term **sex inheritance** to describe this.

All organisms also inherit other features from their parents. You will find out more about this in the next topic.

Activity 7.2.1**Modelling sex inheritance**

In this activity, you will work in pairs and use coloured pieces of string or wire to represent X chromosomes and Y chromosomes.

You will need:

- two containers – such as plastic pots, 45 short pieces of string or wire all the same size and the same colour – these represent X chromosomes, 15 short pieces of string or wire all the same size as the X chromosome strings but a different colour – these represent Y chromosomes

Method

- 1 Put 30 X chromosomes into one of the containers. These represent the chromosomes in 30 egg cells.
- 2 Put 15 X chromosomes and 15 Y chromosomes into the other container. These represent the chromosomes in 30 sperm cells.
- 3 One person puts their hand into the first pot – without looking – and takes out a piece of string or wire. Then they take out another piece of string or wire from the other pot.
- 4 Write down the chromosomes in the 'zygote' (fertilised egg) you have produced. Record the result in a table.
- 5 When you have used up all of the 'chromosomes', add up the total numbers of female and male zygotes that you have produced.

Question

- 1 From your results, what can you say about the chance of a baby being a boy or a girl?

Peer assessment

Look at how another pair have recorded their results.

Can you easily understand their results table?

Is it as good as yours, or is it better? What makes one table better than another?

Questions

- 3 Is it correct to say that the sex of a baby is determined by its father, not its mother? Explain your answer.
- 4 Explain, in your own words, why the numbers of boy and girl babies born each year is approximately equal.

Summary checklist

- ☐ I can explain what a gamete is, and describe the male and female gametes of humans.
- ☐ I can explain that fertilisation is the fusion of a male and female gamete.
- ☐ I can describe how sex is inherited in humans, in terms of X and Y chromosomes.